

GET GOOD AT HARD THINGS

A SYSTEM FOR ACADEMIC EXCELLENCE
THROUGH DISCIPLINE,
DEPTH, AND DELIBERATE EFFORT



FIND MENTORS.

Learn from those
who've gone further.



START SMALL.

Every expert was
once a beginner.



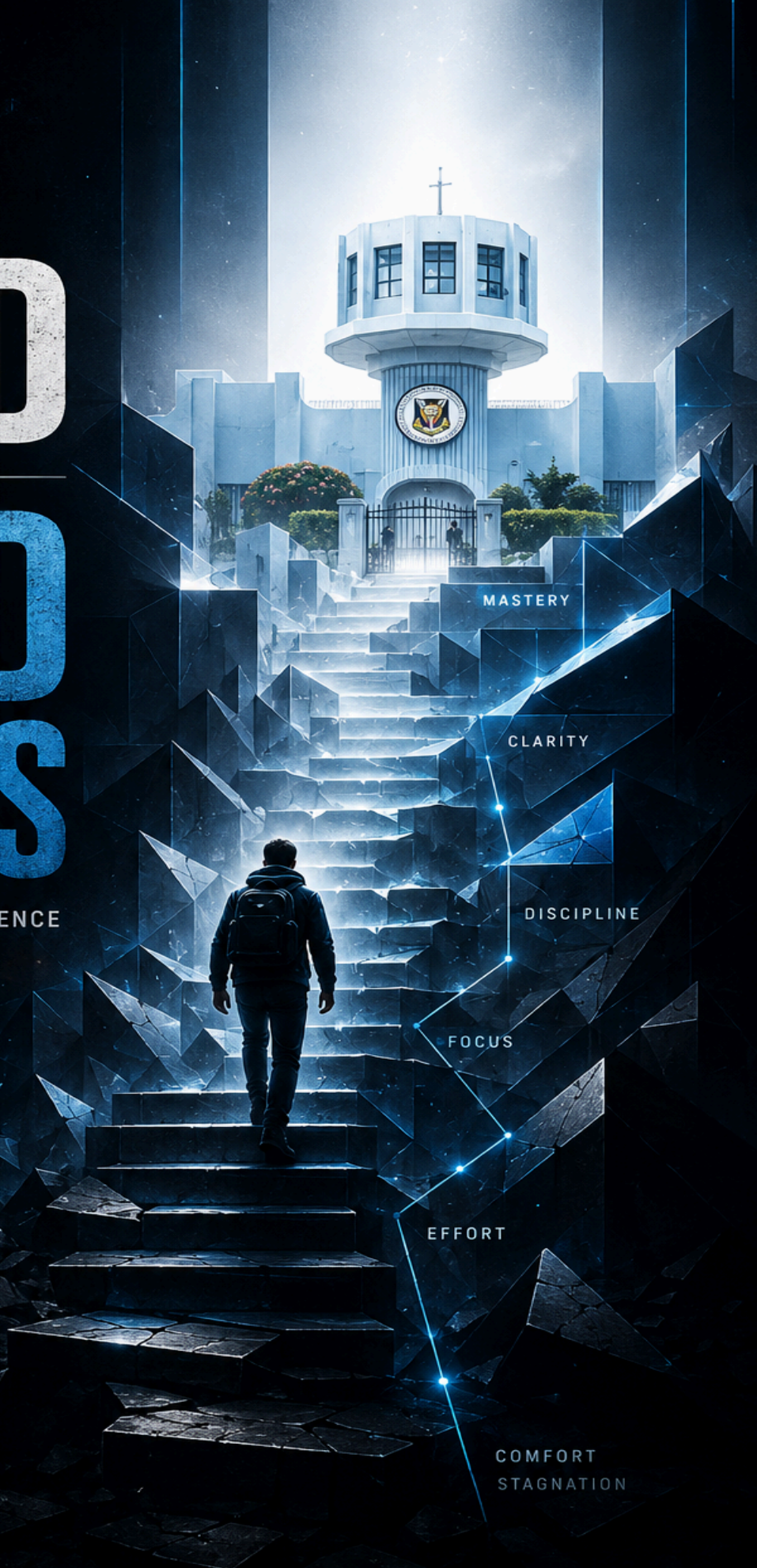
PRACTICE DEEPLY.

Deliberate practice
builds lasting mastery.



COMPOUND CONSISTENCY.

Small efforts today.
Extraordinary impact
tomorrow.



DAVID ITOYA

GET GOOD AT HARD THINGS.

A System For Academic Excellence Through Discipline, Depth,
And Deliberate Effort.

by Itoya David, Founder, Scale Group

Intro.....	5
CHAPTER ONE.....	7
WHAT IS REALLY HAPPENING WHEN IT FEELS LIKE YOUR BRAIN IS STRUGGLING TO UNDERSTAND SOMETHING.....	7
The Productive Discomfort Principle.....	8
CHAPTER TWO.....	11
Why Most Students Feel Like They Studied Hard but Still Perform Poorly in Exams.....	11
Why Passive Studying Feels So Good.....	11
The Netflix Problem.....	12
The Illusion of Learning.....	13
Why High-Performing Students Study Differently.....	14
The Comfort Trap.....	14
The Real Goal of Studying.....	15
What This Means for You.....	16
CHAPTER THREE.....	17
Why the Hardest Ways of Studying Are Usually the Ones That Work Best.....	17
The Brain Does Not Grow Through Comfort.....	17
Why Retrieval Strengthens Memory.....	18
The Hidden Mistake Most Students Make.....	19
Why Difficult Studying Produces Better Results.....	19
Spacing: Why Cramming Fails.....	20
Interleaving: Training the Brain to Think.....	21
Cognitive Resistance Creates Academic Strength.....	21
What This Means for You.....	22
CHAPTER FOUR.....	24
Why It Is Becoming Harder and Harder for Students to Focus And How To Fix It.....	24
Your Brain Learns From Context.....	25
The Problem With Modern Study Environments.....	25
The Attention Residue Problem.....	26
Environment Design Is Academic Strategy.....	27
The Single-Task Principle.....	27
The Digital Environment Matters Too.....	28
Why Elite Students Protect Their Attention.....	29
Building Your Study Environment System.....	29
What This Means for You.....	30
CHAPTER FIVE.....	31
Why Some Students Always Seem Organized While Everyone Else Is Constantly Overwhelmed	31
The Brain Is Bad at Holding Open Loops.....	32
Why Systems Reduce Anxiety.....	32

The Calendar Principle.....	33
The Daily Review Habit.....	34
Weekly Planning Prevents Academic Chaos.....	34
Why Cramming Feels Necessary.....	35
The Importance of Sequential Execution.....	36
Systems Create Consistency.....	36
Building Your Execution System.....	37
What This Means for You.....	37
CHAPTER SIX.....	39
What to Do When You Feel Like You Have Already Fallen Too Far Behind.....	39
Falling Behind Is Normal.....	40
The Spiral Problem.....	40
Step One: Stop the Spiral Immediately.....	41
Step Two: Triage Ruthlessly.....	42
Critical.....	42
Recoverable.....	42
Droppable.....	42
Step Three: Build a 48-Hour Recovery Plan.....	43
The Reset Principle.....	44
When Confidence Collapses.....	44
Why High-Performing Students Recover Faster.....	45
The Real Goal of Recovery.....	46
What This Means for You.....	47
Before You Leave.....	48
One More Thing Before You Go.....	49

Intro

I finished my first year with a perfect CGPA; all As, no Bs.

Over time, I have read countless books on learning, performance, psychology, and academic excellence so that you do not have to. And across all of them, one pattern appears again and again: the highest performers consistently choose what is difficult over what is comfortable in the moment of learning; and they keep making that choice over time.

That single idea, if you truly understand it and practice it, can completely transform your academic journey.

Research has shown over and over again that students become academic high achievers not through long, passive study sessions, but through short, intense periods of focused effort. Techniques like retrieval practice, spaced repetition, and interleaving; methods that are far more difficult and uncomfortable than simply rereading notes; consistently produce much better results. The truth is, the students who rise above average are usually the ones willing to do the hard, mentally demanding work that others avoid. Hard things are what separate the stars from everybody else.

So as you read this book and think deeply about the ideas in it, I want you to ask yourself one honest question: *Am I a star, or am I everybody else?*

Your answer to that question will shape how you approach this book. A star does not just read and move on. A star studies the ideas, applies them, practices them repeatedly, and keeps showing up even when it becomes difficult or uncomfortable. That is what produces real results. That is what separates exceptional people from the crowd.

Everything you are about to read in this book is practical. These are not ideas borrowed from motivational speeches or theories that sound good only on paper. The principles in this book are drawn from methods that have helped students improve their academic performance, study more effectively, and perform at a far higher level than they previously thought possible.

If applied consistently, the strategies in this book can completely change the way you learn. They can help you move from struggling to keep up to studying with clarity and confidence. They can help you move from average grades to academic excellence — from barely passing to earning distinctions, from a third class to a second class upper, or even to a first class.

But there is one problem that stops most students from experiencing real improvement: they consume information without applying it. They read books, watch videos, highlight notes, and feel productive, yet never truly practice what produces results.

This book will not change your academic life simply because you read it. It will only work if you use it.

My hope is that you will approach these pages differently from the average student. That you will not just admire the ideas, but practice them. That you will be disciplined enough to study deliberately, even when it feels difficult, repetitive, or uncomfortable.

Because in academics, the students who improve the most are rarely the most naturally gifted. More often, they are the ones willing to consistently do the hard things others avoid.

If you commit yourself to applying what you learn here, your results will eventually speak for themselves.

Good luck.

CHAPTER ONE

WHAT IS REALLY HAPPENING WHEN IT FEELS LIKE YOUR BRAIN IS STRUGGLING TO UNDERSTAND SOMETHING.

Imagine you are learning to ride a bicycle for the first time.

At first, it feels unstable. Your balance is off, your legs are unsure, and every few seconds you feel like you're about to fall. You might even laugh nervously or get frustrated because nothing feels smooth. In that moment, it is tempting to conclude, *"Maybe I'm just not good at this."*

But the truth is simpler: that wobbling is not a sign of failure. It is the process of learning itself.

Now compare two children. One child only watches videos of people riding bikes. They study it over and over, they understand how it works, and they can even explain it perfectly. The other child gets on the bike, falls, adjusts, tries again, falls less, and slowly begins to ride.

Only one of them will actually learn how to ride.

That difference is the entire idea behind this chapter.

Learning, especially at university level, works the same way. The "wobble," the confusion when you try to recall information, the struggle when answering difficult questions, the mental strain when trying to connect ideas; is not a warning sign that you are failing. It is the exact moment your brain is adapting.

Most students misunderstand this. They interpret difficulty as a signal to stop, switch to something easier, or go back to rereading familiar notes. But the students who eventually excel are the ones who stay on the "bicycle" long enough for balance to develop.

This chapter is about learning to trust that wobble.

Because in academics, just like in cycling, stability only comes *after* instability.

The Productive Discomfort Principle

The Productive Discomfort Principle states that cognitive discomfort during learning is not a signal of failure, but a reliable indicator that meaningful learning is occurring. It holds that academic growth is not produced by ease or repetition alone, but by sustained engagement with appropriately difficult material that challenges current understanding.

According to this principle, a student should not primarily evaluate their learning by how comfortable or fluent the process feels, but by whether the process requires effortful thinking, retrieval, and reconstruction of knowledge. In this view, confusion, mental resistance, and struggle are not obstacles to learning; they are conditions through which learning is developed.

A student applying this principle does not ask, *“Do I understand this easily?”* Instead, they ask, *“Is this level of difficulty forcing me to think in a way that improves my understanding?”*

The principle therefore reframes academic identity away from performance-based self-judgment (such as grades or perceived intelligence) and toward process-based evaluation; specifically, the quality and consistency of engagement with cognitive difficulty over time.

What this means for you is simple: **if studying feels hard, slow, or mentally uncomfortable, that is not a sign that you are failing; it is often a sign that you are actually learning.** Your progress will not come from what feels easy or familiar, but from consistently engaging with work that challenges your thinking and forces you to struggle a little before you understand. Instead of asking whether something feels easy or smooth, you should be asking whether it is pushing your brain to think harder and build real understanding.

In other words, your growth in school will depend less on how smart you feel in the moment and more on how willing you are to stay with difficult thinking long enough for it to make you better.

To a student practicing this principle, mental discomfort during studying is not a sign of failure, but one of the clearest signs that real learning is taking place. It means you stop measuring your academic ability by how easy studying feels and start measuring it by whether you are engaging with the kind of difficulty that actually produces growth.

A student operating under this principle no longer asks, “Am I smart enough for this course?” Instead, they ask, “Am I studying in a way that challenges my brain to grow?”

That shift changes everything.

Most students are naturally drawn toward comfort. We prefer rereading notes instead of testing ourselves because rereading feels smooth and familiar. We prefer highlighting textbooks instead of solving difficult questions because it creates the feeling of productivity without the mental strain. But the problem is that familiarity can easily be mistaken for mastery.

Real learning rarely feels comfortable in the moment.

Think about a first-year student who joins the gym for the first time. If every workout feels easy, if there is no tension in the muscles, no resistance, no exhaustion, then very little growth is happening. Muscle develops through strain. The discomfort is not evidence that the workout is failing; it is evidence that the body is adapting.

Learning works the same way.

When you struggle to recall information without checking your notes, when you wrestle with difficult practice questions, when you force yourself to connect ideas across multiple topics, your brain experiences resistance. That resistance is what strengthens understanding, memory, and problem-solving ability.

This is why the highest-performing students often study in ways that feel harder, not easier. They do not avoid cognitive difficulty; they deliberately engage with it. While the average student studies to feel comfortable, the exceptional student studies to become capable.

This idea is strongly reflected in the work of learning educator Justin Sung, who emphasizes that effective learning should involve meaningful cognitive effort. If your study sessions are always passive, smooth, and mentally effortless, there is a good chance you are optimizing for the feeling of studying rather than genuine understanding.

You likely already understand this principle outside academics. Nobody expects to become good at football by merely watching highlights. Nobody expects to become skilled at playing an instrument by simply rereading lyrics. Skill is built through repetition, mistakes, correction, and sustained difficulty.

Yet many students enter university expecting studying to feel easy all the time. The moment they encounter confusion, mental fatigue, or difficulty recalling information, they assume something is wrong with them. In reality, those moments are often where the deepest learning begins.

The students who eventually graduate with first class are not always the students with the highest natural intelligence. Very often, they are the students who become comfortable with productive discomfort. They learn not to run from difficult thinking, but to remain inside it long enough for growth to happen.

Learning only occurs in the zone where performance is being stretched. Identifying with the struggle; rather than the result; is what sustains the behavior long enough to produce elite outcomes.

CHAPTER TWO

Why Most Students Feel Like They Studied Hard but Still Perform Poorly in Exams

Imagine a student preparing for an important exam.

They sit at their desk for six hours straight. Their notebook is open. Their textbook is covered in highlighter. They reread the same pages repeatedly, nodding occasionally as the information begins to look more familiar.

By the end of the night, they feel productive.

After all, they studied for six hours.

Then the exam comes.

Suddenly, the information they “knew” becomes difficult to recall. Questions that looked familiar the night before now feel strangely unfamiliar under pressure. Their brain freezes. They recognize the topic when they see it, but they cannot reconstruct the answer clearly enough to write it.

After the exam, they say something many students say:

“I actually read for this exam. I don’t know what happened.”

What happened is simpler than most students realize.

They confused exposure with learning.

And that confusion destroys the academic performance of thousands of students every year.

Why Passive Studying Feels So Good

Passive studying includes activities like:

- rereading notes repeatedly

- highlighting textbooks excessively
- watching lectures without testing understanding
- copying information word-for-word
- reviewing material without trying to recall it independently

The reason these methods are popular is because they feel comfortable.

The brain likes familiarity. When you repeatedly see the same information, it begins to feel easier to process. You start recognizing terms quickly. Concepts look familiar. Notes appear clearer each time you reread them.

This creates the feeling of progress.

But feeling like you know something is not the same as being able to use it.

A student can read a physiology definition ten times and still fail to explain it without their notes open.

A student can watch an entire mathematics tutorial and still struggle to solve a similar problem independently.

A student can highlight an entire textbook and still blank out in the exam hall.

Because recognition is not mastery.

The Netflix Problem

Think about how easily you recognize scenes from a movie you have watched many times.

You may know the dialogue before the characters even speak. You recognize faces instantly. You can predict what happens next because the material is familiar.

But imagine someone suddenly pauses the movie halfway and says:

“Now recreate the entire scene yourself from memory.”

Most people would struggle immediately.

Why?

Because recognition is easier than reconstruction.

Watching a movie repeatedly creates familiarity, not production ability.

Studying works the same way.

When students reread notes, the brain begins recognizing information more efficiently. But exams do not primarily test recognition. They test retrieval.

The exam hall does not ask:

“Does this look familiar?”

It asks:

“Can you produce this information independently?”

That is a completely different skill.

And this is where most students fail.

The Illusion of Learning

One of the most dangerous experiences in academics is the illusion of understanding.

This happens when information feels easy simply because you have encountered it multiple times.

A student rereads a chapter three times and concludes:

“I know this.”

But the real test is simple:

Close the book.

Remove the notes.

Take a blank sheet of paper.

Now explain the concept from memory.

Suddenly, the “understanding” disappears.

This is why many students are shocked by poor exam results. They mistake familiarity for mastery and exposure for competence.

Real learning is proven by what you can retrieve, explain, connect, and apply without support.

Not by how smooth rereading feels.

Why High-Performing Students Study Differently

High-performing students understand something average students often miss:

The goal of studying is not to see information.

The goal is to build the ability to retrieve and use information independently.

That changes everything.

Instead of rereading passively, they test themselves actively.

Instead of highlighting endlessly, they attempt difficult questions.

Instead of consuming information repeatedly, they force themselves to reconstruct it from memory.

This approach feels harder.

And that is precisely why it works.

When your brain struggles to retrieve information, it strengthens the pathways associated with that information. The effort itself becomes part of the learning process.

This is one of the key ideas emphasized by learning educator Justin Sung: effective studying is not about maximizing comfort or exposure, but about maximizing meaningful cognitive engagement.

The students who improve most are usually not the students spending the most time passively looking at material.

They are the students spending the most time thinking.

The Comfort Trap

Most students unknowingly optimize for comfort.

They choose study methods that reduce mental strain:

- rereading instead of retrieval
- watching instead of practicing
- recognizing instead of recalling
- consuming instead of producing

Why?

Because difficult studying can feel discouraging.

The moment students test themselves and cannot remember something quickly, they panic. They interpret the struggle as evidence that they are weak academically.

So they return to methods that feel smoother.

But smooth studying is often deceptive studying.

The methods that feel easiest during learning frequently produce the weakest long-term retention.

Meanwhile, the methods that feel mentally demanding often produce the strongest understanding over time.

This is why passive studying is dangerous.

It gives students emotional comfort without guaranteed academic competence.

The Real Goal of Studying

The purpose of studying is not to become familiar with information.

The purpose is to become capable with information.

There is a difference.

A familiar student says:

"I've seen this before."

A capable student says:

"I can explain this, apply this, and retrieve this under pressure."

University rewards capability, not familiarity.

And capability is built through active engagement, not passive exposure.

This means real studying will sometimes feel slower, harder, and less satisfying in the moment.

But those uncomfortable moments are often where the deepest learning is taking place.

What This Means for You

If you want to improve academically, you must stop using comfort as the measure of effective studying.

A study session should not be judged by how long you sat at your desk or how familiar your notes became.

It should be judged by whether your brain was forced to think, retrieve, connect, and reconstruct knowledge actively.

The question is no longer:

“Did I look at the material?”

The question becomes:

“Could I produce it independently if the material disappeared?”

That is the standard real learning requires.

And once you understand this, your entire approach to studying begins to change.

CHAPTER THREE

Why the Hardest Ways of Studying Are Usually the Ones That Work Best

Imagine two students preparing for the same examination.

The first student spends the evening reading lecture notes repeatedly. They underline important definitions, highlight formulas, and carefully organize their pages. By the end of the night, everything looks familiar. The material feels easier to recognize each time they see it.

The second student studies differently.

After briefly reviewing the material, they close their notes and begin testing themselves. They try to explain concepts from memory. They solve practice questions without checking the answers immediately. They struggle to remember certain points. They get some questions wrong. They pause, think harder, and try again.

By the end of the session, the second student feels mentally exhausted.

If you asked both students who studied better, the first student would probably feel more confident.

Why?

Because passive studying feels smoother.

But smoothness is not always a sign of learning.

In many cases, the student whose studying felt harder is actually building stronger understanding.

That is because the brain develops through cognitive resistance.

The Brain Does Not Grow Through Comfort

Most students understand this principle physically, but not academically.

Everybody understands that muscles grow through resistance. Nobody expects physical strength to develop without tension, repetition, and strain.

But many students secretly expect learning to happen comfortably.

They expect understanding to feel immediate.

They expect memory to form effortlessly.

And when studying becomes mentally demanding, they assume something is wrong.

But the brain adapts the same way the body does.

Mental strength develops through mental resistance.

This is why effective learning often feels effortful. Your brain is being forced to retrieve information, organize ideas, solve problems, and build stronger neural connections.

That process requires energy.

And anything that requires genuine adaptation usually feels uncomfortable while it is happening.

Why Retrieval Strengthens Memory

Imagine someone asks you for the phone number of an old friend you have not contacted in years.

At first, your brain struggles. You pause. You search mentally. Certain digits appear, disappear, then return again. After a few seconds, you finally remember the number.

That struggle matters.

The act of pulling information out of memory strengthens the memory itself.

This is why retrieval practice is so powerful.

Every time you force your brain to recall information without immediately checking your notes, you strengthen your ability to access that information again later.

This is completely different from rereading.

Rereading exposes you to information.

Retrieval trains you to produce information.

And exams reward production, not exposure.

The Hidden Mistake Most Students Make

Most students spend the majority of their study time trying to put information into their brain.

Very few spend enough time trying to pull information out.

That is the mistake.

Learning is not only about input.

It is about retrieval.

A student may read an anatomy chapter five times and still struggle during the exam because the brain was never trained to retrieve the information independently.

Another student may spend less time reading but more time testing themselves, recalling concepts, answering questions, and reconstructing diagrams from memory.

That student usually performs better because their studying resembled the conditions of the exam itself.

This is one of the central ideas emphasized by learning educator Justin Sung: high-quality studying requires active cognitive processing, not passive exposure. Deep learning happens when students are forced to manipulate, retrieve, connect, and restructure knowledge mentally.

The brain remembers what it works hard to retrieve.

Why Difficult Studying Produces Better Results

One of the strangest truths about learning is this:

The study methods that **feel** most effective are often the **least effective**.

And the methods that feel difficult are often the most powerful.

Why?

Because easy studying can create an illusion of mastery.

When students reread notes, everything looks familiar, so they assume understanding exists. But when they are forced to recall information independently, the gaps become visible immediately.

That discomfort is valuable.

It reveals weakness before the exam reveals it publicly.

High-performing students understand this.

They deliberately create situations where their understanding is tested before the real test arrives.

They quiz themselves.

They attempt difficult questions.

They explain concepts without notes.

They simulate exam conditions early.

This approach feels mentally heavier, but it produces stronger learning because the brain is being trained actively instead of entertained passively.

Spacing: Why Cramming Fails

Another major mistake students make is believing that long, exhausting study marathons produce better retention.

But the brain does not learn best through endless exposure in one sitting.

It learns through repeated retrieval across time.

This is why students often forget material they crammed the night before an exam. The information was temporarily accessible, but it was never strengthened through repeated reconstruction over longer intervals.

Spacing changes this.

When you revisit information after some forgetting has occurred, retrieval becomes slightly harder. That difficulty forces the brain to work again, which strengthens memory further.

In other words:

Forgetting slightly before reviewing is not a problem.

It is part of the learning process.

This is why students who review material consistently across weeks usually outperform students who study intensely only once.

Learning is built through repeated cognitive effort over time.

Not through panic-driven memorization sessions.

Interleaving: Training the Brain to Think

Many students study one topic repeatedly until it feels comfortable before moving to another topic.

This feels organized, but it can create weak flexibility.

Real exams rarely present questions in perfectly separated categories. They require students to recognize which concept applies in unfamiliar situations.

This is where interleaving becomes powerful.

Interleaving means mixing different topics or problem types together during study.

At first, this feels confusing because the brain cannot rely on repetition patterns easily. It must constantly identify, compare, and adapt between ideas.

That extra difficulty improves long-term understanding.

The brain becomes better at selecting the correct approach independently instead of merely repeating memorized procedures.

Again, productive difficulty creates stronger learning.

Cognitive Resistance Creates Academic Strength

The students who improve most academically are not necessarily the students studying the longest hours.

Often, they are the students whose study sessions contain the highest quality mental effort.

They retrieve instead of reread.

They test instead of merely review.

They struggle before checking the answer.

They revisit material across time instead of cramming once.

They deliberately place their brain under controlled cognitive resistance.

And over time, that resistance changes them.

What begins as slow, difficult, frustrating effort eventually becomes fluency, speed, and confidence.

Just like physical training, the discomfort comes first.

The strength comes later.

What This Means for You

If you truly want to improve academically, you must stop treating studying as passive exposure to information.

Studying is cognitive training.

Your brain develops through effortful retrieval, problem-solving, reconstruction, and repeated engagement with difficulty.

This means effective studying will not always feel pleasant.

Sometimes it will feel slow.

Sometimes it will expose how much you do not know.

Sometimes it will frustrate you.

But those moments are often where the deepest learning is happening.

The goal is no longer to make studying feel easy.

The goal is to make your brain stronger.

And strength is always built through resistance.

CHAPTER FOUR

Why It Is Becoming Harder and Harder for Students to Focus And How To Fix It

Imagine two students preparing for the same examination.

The first student studies on their bed with their phone beside them. Music is playing in the background. WhatsApp notifications appear every few minutes. Instagram opens “just for two minutes.” Multiple browser tabs are open at the same time — lecture notes, YouTube, Twitter, messages, and unrelated searches competing constantly for attention.

The second student studies differently.

Before beginning, they place their phone in another room(or just lock it in your desk; the goal is to make it hard for you to get it). Their desk contains only the materials needed for one task. Notifications are disabled. Their study location is quiet and consistent. The environment itself communicates one message to the brain:

“We are here to work.”

Now imagine both students repeating their habits every day for an entire semester.

Who do you think develops deeper concentration?

Who do you think finishes more meaningful work?

Who do you think enters exams with stronger understanding?

The difference is not intelligence.

It is environment.

Most students believe focus is something they must force through motivation or discipline alone. But in reality, environment shapes behavior far more than people realize.

Your surroundings are either helping your concentration or quietly destroying it.

There is rarely a neutral middle ground.

Your Brain Learns From Context

One of the most overlooked truths about studying is that the brain forms associations with environments.

If you repeatedly watch movies in bed, your brain begins associating the bed with relaxation.

If you repeatedly scroll social media at your desk, your brain begins associating the desk with distraction.

And if you consistently perform deep, focused work in a specific location, your brain gradually learns to enter concentration more quickly when you arrive there.

This is why many high-performing students study in consistent locations.

Not because the location itself is magical, but because consistency conditions the brain.

Over time, the environment becomes a cognitive trigger.

The moment they sit down, the brain already understands the assignment:

Focus starts here.

The Problem With Modern Study Environments

Most students are trying to study in environments designed for distraction.

The modern digital environment constantly competes for attention:

- notifications
- messages
- short videos
- trending topics
- endless scrolling
- multiple tabs
- background entertainment

Every interruption may seem small, but attention fragmentation has consequences.

Deep understanding requires sustained cognitive engagement.

But distraction repeatedly breaks the brain out of that process before understanding fully develops.

Many students believe they can multitask effectively:

- studying while chatting
- reading while watching videos
- revising while checking notifications

But attention does not truly multitask.

It switches.

And every switch carries a cognitive cost.

The brain wastes energy constantly reorienting itself between tasks instead of going deeper into one task long enough for meaningful learning to occur.

The Attention Residue Problem

Imagine trying to solve a difficult mathematics problem while someone interrupts you every two minutes asking unrelated questions.

Even after the interruption ends, part of your attention remains attached to what distracted you.

This is called attention residue.

Your body may still be sitting at the desk, but your mind is partially elsewhere.

This is one reason social media destroys study quality so effectively. Even brief interruptions leave fragments of attention behind.

A student may technically study for five hours while only producing two hours of real concentration.

This creates one of the biggest illusions in academics:

time spent studying is mistaken for quality studying.

But high performance is not built mainly through time.

It is built through depth of focus.

Environment Design Is Academic Strategy

High-performing students often do not rely entirely on willpower.

Instead, they reduce friction around good behavior and increase friction around distraction.

In other words, they design environments that make focus easier.

This can look very simple:

- phone placed in another room
- only one task visible at a time
- quiet locations away from unnecessary movement
- study materials already organized before sessions begin
- distracting tabs closed completely
- consistent study locations used repeatedly

These details may seem small, but repeated small advantages create enormous long-term differences.

The environment either protects concentration or leaks it constantly.

The Single-Task Principle

One of the most important focus rules is simple:

Only one cognitive task should be visible at a time.

The brain performs best when attention is directed fully toward a single objective.

But many students create visually overloaded environments:

- multiple books open
- random tabs everywhere
- phone notifications appearing
- unrelated assignments scattered across the desk

Even if those distractions are not being used directly, the brain still processes their presence partially.

This creates unnecessary cognitive load.

A cleaner environment produces clearer thinking.

When it is time to study physiology, only physiology should exist in front of you.

When it is time to solve mathematics problems, everything unrelated should disappear temporarily.

Focus strengthens when attention stops competing with itself. Declutter.

The Digital Environment Matters Too

A study environment is not only physical.

It is digital as well.

Many students create digital chaos without realizing it:

- files scattered randomly
- lecture materials disorganized
- multiple note-taking systems
- forgotten deadlines
- inconsistent storage methods

This creates hidden stress because the brain wastes energy searching, remembering, and reorganizing constantly.

Organization reduces cognitive friction.

Simple systems matter:

- course materials organized by subject
- notes arranged by week or topic
- one consistent note-taking system
- scheduled times for email checking
- no unnecessary tabs during study sessions

The goal is not perfection.

The goal is reducing unnecessary mental noise.

Your brain should spend its energy learning, not searching.

Why Elite Students Protect Their Attention

Attention is one of the most valuable academic resources you possess.

Once attention becomes fragmented repeatedly, deep thinking becomes difficult.

And deep thinking is required for:

- understanding difficult concepts
- solving unfamiliar problems
- building long-term memory
- making connections between ideas

This is why high-performing students often appear “boring” during serious academic periods.

They protect their attention aggressively.

Not because they hate enjoyment.

But because they understand that every environment trains the brain in a particular direction.

A distracted environment trains distracted thinking.

A focused environment trains focused thinking.

Building Your Study Environment System

Your environment should communicate clarity, not chaos.

A strong study environment usually includes:

- a consistent study location
- minimal distractions
- phone absent or inaccessible
- water available
- paper and pen nearby for active recall
- organized digital materials
- only one visible task at a time

Most importantly, your environment should make deep work easier than distraction.

Because discipline becomes significantly easier when the environment supports the behavior you want.

What This Means for You

If your study environment constantly destroys your attention, your academic performance will eventually suffer no matter how intelligent you are.

Focus is not only psychological.

It is environmental.

This means academic improvement is not just about studying harder. It is also about designing conditions that allow concentration to happen consistently.

Your surroundings are shaping your thinking every single day.

The question is whether they are shaping it toward distraction or toward depth.

Because students rarely rise above the environments they repeatedly place themselves inside.

CHAPTER FIVE

Why Some Students Always Seem Organized While Everyone Else Is Constantly Overwhelmed

Imagine two students beginning a new semester.

The first student enters the semester with good intentions but no real system. Assignment deadlines remain in their head. Test dates are remembered vaguely. Study sessions happen “when there is time.” Some days are productive, many are not. Important tasks are postponed until pressure becomes unbearable.

At first, this feels manageable.

Then the semester accelerates.

Multiple deadlines arrive at once. Notes begin piling up. Missed readings accumulate quietly. Anxiety increases. The student constantly feels busy, yet strangely behind at the same time.

Now compare this to another student.

This student records every deadline immediately. They review their schedule each morning briefly. They plan each week intentionally. Their studying is distributed across time instead of compressed into panic sessions. Tasks are captured externally instead of being carried mentally.

By the middle of the semester, one student feels overwhelmed.

The other feels organized.

The difference is not motivation.

It is systems.

Most students underestimate how much academic success depends on execution structure rather than intelligence alone.

A strong student is rarely just “hardworking.”

More often, they are organized enough to direct their effort consistently over time.

The Brain Is Bad at Holding Open Loops

Many students rely heavily on memory to manage their academic life:

- “I’ll remember that assignment.”
- “I’ll study that later.”
- “I know the test is sometime next week.”
- “I’ll organize everything tomorrow.”

This creates invisible cognitive stress.

Every unfinished obligation occupies mental space.

Psychologists sometimes refer to this as an “open loop” — unresolved tasks competing silently for attention in the background.

The brain is excellent at thinking.

It is terrible at storing large amounts of unorganized responsibility reliably.

This is why students often feel mentally exhausted before they even begin studying. Their attention is overloaded by unfinished reminders, forgotten obligations, and unclear priorities.

The solution is simple:

Stop using your brain as storage.

Use it for thinking instead.

Why Systems Reduce Anxiety

One of the biggest sources of academic stress is uncertainty.

Students become anxious not only because work exists, but because they do not know:

- what must be done first
- how much work actually exists
- when deadlines are approaching
- whether they are forgetting something important

A strong execution system reduces this uncertainty.

The moment responsibilities are captured externally; on a calendar, planner, or task system; the brain relaxes slightly because it no longer needs to carry everything mentally.

Clarity reduces stress.

Chaos increases it.

This is why high-performing students often appear calmer during difficult academic periods. It is not necessarily because they have less work.

It is because their work is structured.

The Calendar Principle

Every important academic commitment should be captured immediately.

Not later.

Not eventually.

Immediately.

Exam dates.

Submission deadlines.

Presentations.

Group meetings.

Clinical postings.

Laboratory sessions.

Everything.

The moment information enters your life, it should enter your system.

Why?

Because memory is unreliable under stress.

Many students trust themselves too much in this area. They assume they will remember everything naturally until multiple responsibilities collide at the same time.

At that point, forgotten obligations become emergencies.

A calendar prevents this.

Your calendar is not merely a scheduling tool.

It is an external memory system.

And external systems are more reliable than mental guessing.

The Daily Review Habit

One of the simplest high-performance habits is also one of the most powerful:

A brief daily review.

Every morning, take two minutes to look at:

- deadlines
- scheduled work
- upcoming responsibilities
- unfinished tasks

This small action creates awareness before the day begins.

Without review, students drift reactively through the day responding emotionally to whatever feels urgent.

With review, students operate intentionally.

They know what matters before distraction arrives.

A strong academic life is rarely built through dramatic moments of motivation.

It is usually built through small repeated acts of organization.

Weekly Planning Prevents Academic Chaos

Many students only think seriously about school when pressure becomes extreme.

This is dangerous.

High-performing students think ahead.

A weekly planning session — even ten minutes every Sunday — can dramatically reduce academic instability.

During this session, students review:

- upcoming deadlines
- difficult subjects needing extra time
- assignments approaching completion
- study sessions for the week
- possible overload points

This creates proactive behavior instead of reactive survival.

Students who plan weekly rarely eliminate stress completely.

But they usually prevent avoidable chaos.

And preventing chaos matters academically because panic destroys cognitive performance.

Why Cramming Feels Necessary

Students often cram because they lack execution systems earlier in the semester.

Work accumulates silently.

Avoidance delays engagement.

Then eventually, urgency arrives all at once.

Cramming is usually not a time-management problem alone.

It is often a delayed-decision problem.

Tasks postponed repeatedly eventually combine into overwhelming pressure.

Strong execution systems reduce the need for emergency studying because academic effort is distributed more consistently across time.

This is one reason elite students often appear less stressed during examinations.

They are not always smarter.

They simply started earlier.

The Importance of Sequential Execution

One major mistake students make is trying to solve everything mentally at the same time.

This creates paralysis.

When ten unfinished tasks compete simultaneously for attention, the brain struggles to prioritize clearly.

Execution becomes emotionally overwhelming.

The solution is sequential focus.

Choose one task.

Complete it.

Then move to the next.

High-performing students rarely attempt to mentally carry their entire semester at once.

They reduce complexity into executable next actions.

This creates movement.

And movement reduces anxiety.

Systems Create Consistency

Motivation fluctuates constantly.

Some days you will feel inspired.

Many days you will not.

Students who depend entirely on motivation often become inconsistent because emotions change faster than responsibilities do.

Systems solve this problem.

A calendar still exists when motivation disappears.

A study schedule still exists when mood changes.

A planning routine still functions when energy is low.

This is why systems outperform motivation over long periods of time.

Motivation creates occasional intensity.

Systems create sustained execution.

And sustained execution is what academic excellence requires.

Building Your Execution System

Your execution system does not need to be complicated.

It simply needs to reduce confusion and increase clarity.

A strong system usually includes:

- a digital or physical calendar
- immediate deadline capture
- daily review habits
- weekly planning sessions
- scheduled study blocks
- organized task tracking
- external storage of responsibilities

Most importantly, no important obligation should exist only inside your memory.

Every responsibility should live somewhere visible and reliable.

What This Means for You

Academic success is not only about understanding information.

It is also about managing yourself effectively across time.

Without systems, even intelligent students eventually become overwhelmed by accumulated disorder.

But when responsibilities are captured clearly, planned intentionally, and executed sequentially, academic life becomes far more stable.

The goal is not to become perfect.

The goal is to become organized enough that your energy is spent learning instead of constantly recovering from preventable chaos.

Because high-performing students are rarely carrying less responsibility.

They are simply carrying it more systematically.

CHAPTER SIX

What to Do When You Feel Like You Have Already Fallen Too Far Behind

Imagine a student sitting alone late at night staring at unfinished work.

Three assignments are overdue.

Lecture notes have piled up for weeks.

An upcoming test is approaching quickly.

The student opens their laptop, looks at everything they have missed, feels overwhelmed almost immediately, and then quietly closes it again.

Not because they are lazy.

But because the gap now feels emotionally unbearable.

At that moment, many students stop studying entirely.

They avoid checking school portals.

They ignore messages from classmates.

They stop attending lectures consistently.

And slowly, one missed task becomes a collapsed semester.

This is one of the most dangerous moments in academic life.

Not the moment a student falls behind.

But the moment they begin identifying with the failure emotionally.

Because once students conclude:

"I am too far behind,"

they often stop taking the actions that could have recovered the situation.

That is why every high-performing student needs something most students never build:

A failure recovery system.

Falling Behind Is Normal

One of the biggest misconceptions students carry is the belief that successful students never struggle academically.

This is false.

Even excellent students fall behind sometimes.

They get overwhelmed.

They mismanage time occasionally.

They underestimate workloads.

They experience burnout.

They perform poorly on tests unexpectedly.

The difference is not that elite students avoid all failure.

The difference is that they recover faster.

Average students often turn temporary setbacks into identity crises.

High-performing students treat setbacks as system problems requiring adjustment.

That distinction matters enormously.

The Spiral Problem

When students fall behind, they usually make the same mistake:

They try to emotionally process the entire situation at once.

They think about:

- every missed lecture
- every overdue assignment
- every upcoming exam

- every mistake they made
- every possible consequence

The brain becomes overloaded.

And overload often produces paralysis.

This creates what can be called the academic spiral.

The student feels overwhelmed → avoids work temporarily → falls further behind → feels worse → avoids work again.

Over time, the emotional weight becomes heavier than the academic workload itself.

This is why recovery must become mechanical instead of emotional.

Step One: Stop the Spiral Immediately

The first rule of recovery is simple:

Do not disappear emotionally into the problem.

Within 24 hours of recognizing that you are behind, stop and assess the situation clearly.

Not emotionally.

Practically.

Sit down and write out every outstanding obligation:

- assignments
- readings
- lectures
- quizzes
- projects
- deadlines

Everything.

Why?

Because vague problems feel bigger than defined problems.

The brain panics most when responsibility feels unclear and shapeless.

Writing things down transforms emotional chaos into visible reality.

And visible reality is easier to solve.

Step Two: Triage Ruthlessly

Once everything is visible, the next step is prioritization.

Not all academic tasks carry equal importance.

This is where many students fail.

They attempt to recover everything equally at once and collapse under the pressure.

Instead, classify tasks into three categories:

Critical

Tasks that directly affect grades significantly:

- exams
- major assignments
- graded submissions

These receive immediate attention.

Recoverable

Tasks that still matter but can be compressed:

- practice problems
- partial readings
- review materials

These can be handled strategically.

Droppable

Low-impact tasks:

- optional readings
- supplemental material
- nonessential perfectionism

These are removed without guilt.

This is important because overwhelmed students often waste energy trying to preserve perfection instead of restoring functionality.

But recovery is not about perfection. It is about stabilization.

Step Three: Build a 48-Hour Recovery Plan

One of the biggest mistakes students make after falling behind is trying to “fix their entire life” overnight.

This almost always fails.

The brain performs better with manageable targets.

Instead of focusing on the entire semester, focus only on the next 48 hours.

Ask:

“What can realistically be completed within the next two days?”

Then schedule those actions specifically.

Not vaguely.

Specifically.

For example:

- 9:00–10:30 AM — complete physiology assignment
- 11:00–12:00 PM — review lecture slides
- 2:00–3:30 PM — solve practice questions

Clarity creates movement.

And movement interrupts paralysis.

The goal is not immediate perfection.

The goal is regaining momentum.

Because momentum changes psychology quickly.

The Reset Principle

One of the most important ideas in academic recovery is this:

You do not need to “catch up perfectly” to recover successfully.

You only need to restart effectively from the current moment forward.

Many students become psychologically trapped trying to erase the past completely.

But academic recovery rarely works that way.

A student who falls three weeks behind does not need to emotionally carry three weeks forever.

They need to execute competently now.

This is the Reset Principle:

There is no perfect recovery point. There is only the next correct action.

Once students understand this, recovery becomes lighter psychologically.

The focus shifts from:

“How do I erase my mistakes?”

to:

“What is the next useful action available right now?”

That shift restores agency.

When Confidence Collapses

Sometimes the problem is not workload.

Sometimes the problem is belief.

A poor exam result can damage a student’s confidence deeply. After repeated struggles, many students begin attaching academic outcomes to personal identity:

“Maybe I’m not intelligent enough.”

This is dangerous because identity-based thinking reduces persistence.

Students stop experimenting.

Stop adjusting.

Stop trying consistently.

But poor outcomes should be treated as information, not definitions.

A bad result does not automatically mean you are incapable.

It may simply reveal:

- weak retrieval practice
- poor spacing
- ineffective study methods
- lack of understanding in a specific area
- inconsistent execution

The correct response is process analysis.

Not self-condemnation.

Ask:

- Did I actively retrieve information?
- Did I test myself repeatedly?
- Did I study across time?
- Did I truly understand the material or merely reread it?

If the process was flawed, improve the process.

If the process was strong but gaps still existed, identify the weak areas specifically and rebuild them carefully.

Failure is data.

And data is useful.

Why High-Performing Students Recover Faster

High-performing students are not emotionally destroyed every time something goes wrong academically.

Not because failure feels good.

But because they understand that setbacks are part of long-term performance development.

They separate:

- outcome from identity
- mistakes from worth
- setbacks from permanence

This allows them to recover faster because they spend less time emotionally collapsing and more time adjusting systems.

They ask:

“What failed?”

instead of:

“What is wrong with me?”

That question protects momentum.

And momentum matters academically because consistency over time is more important than occasional perfection.

The Real Goal of Recovery

The goal of recovery is not emotional comfort.

The goal is restored functionality.

You do not need to feel fully motivated before restarting.

You do not need to feel confident before taking action.

Very often, action rebuilds confidence — not the other way around.

This is important because many students wait emotionally before restarting academically.

That delay usually deepens the problem.

Movement comes first.

Confidence often follows later.

What This Means for You

At some point in your academic journey, you will likely fall behind.

You will make mistakes.

You may fail tests unexpectedly.

You may experience periods of confusion, exhaustion, or inconsistency.

That does not make you incapable.

It makes you human.

The important thing is not avoiding every setback perfectly.

The important thing is building the ability to recover quickly, adjust intelligently, and continue moving forward without turning temporary struggles into permanent identity conclusions.

Because academic success does not belong only to students who never fail.

Very often, it belongs to students who learn how to restart faster than everyone else.

STUDENT FAQ: ACADEMIC PERFORMANCE & TIME MANAGEMENT

FAQ 1: How do you manage your time balancing study and business; and can one stop wasting time on unproductive activities?

The short answer: you do not balance them. You schedule them.

Balance implies that two things exist in comfortable equilibrium. For most students running a business or side activity alongside academics, this is a fantasy that leads to doing both poorly. What actually works is **intentional compartmentalization**; giving each domain a defined time window and protecting those windows absolutely.

Here is the practical structure:

First, map your non-negotiables. Write down every fixed commitment: lectures, practicals, business meetings, transport time, meals. These are immovable. Everything else is schedulable.

Second, assign your remaining hours to one of three categories:

- Deep academic work (high-focus study blocks)
- Business/productive activity
- Genuine recovery (sleep, exercise, real rest)

Third, and most importantly: **eliminate the grey zone**. The grey zone is the hours that belong to neither category — scrolling, half-watching something, chatting without purpose, staring at your phone between tasks. This is where most students lose two to four hours daily without realising it.

On stopping unproductive behaviour:

The research is clear that willpower alone does not work. What works is **environment design and scheduling specificity**. When your next hour has a written, specific task assigned to it, the probability of drifting drops dramatically. When the next hour is vague, the brain defaults to the path of least resistance; which is always low-value activity.

Practical rules:

- Every morning, the next day's schedule is written the night before
- If you are not in a scheduled work block, you are in scheduled rest; not drift

- Your phone is the single largest source of unproductive time for most students. Treat it as a tool with operating hours, not a permanent extension of your hand

On the business specifically: Your business should fund your degree, not compete with it. Define the minimum viable hours your business requires each week, protect those, and cap them. If the business demands more than that, it is a problem to solve, not a reason to sacrifice academic performance.

FAQ 2: Why do I study hard but fail to get even average on my CA test?

This is the most important question in this entire FAQ, because it exposes the most dangerous illusion in education: the belief that effort equals learning.

The research in *Make It Stick* is unambiguous on this point. **The feeling of studying is not the same as the act of learning.** A student can spend six hours with their notes and books and encode almost nothing permanently. Another student can spend 90 minutes and retain it for months. The difference is not effort. It is method.

Here is what hard studying usually looks like; and why it fails:

- Rereading notes and textbooks repeatedly
- Highlighting passages
- Reviewing summaries written by others
- Sitting with material for long hours in a state of familiarity

All of these activities produce **fluency**; the feeling that you know the material because you recognise it when you see it. But recognition is not recall. In a CA test, you are not shown the answer and asked to confirm it. You are given a question and required to produce the answer from memory. These are entirely different cognitive tasks, and passive study only trains one of them.

What you need to do instead:

Close your notes. Write down everything you can remember about the topic. Check what you missed. Return to it tomorrow and test yourself again. This is called retrieval practice, and it is the single most research-validated study technique available. It feels harder. It produces more errors during practice. That is precisely why it works; your brain encodes deeply only when it is forced to reconstruct, not merely recognise.

Additional diagnostics; ask yourself these questions:

- Did I attend all relevant lectures and understand the material during class, or was I relying entirely on catch-up?
- Did I start preparing for this CA more than a week in advance, or in the final days?
- Did I test myself before the exam, or only review?
- Do I understand the concepts, or have I memorised surface-level points without grasping the underlying logic?

If your honest answer to any of these reveals a gap, that is where to begin. Hard work applied to the wrong method produces consistent failure regardless of the hours invested.

FAQ 3: What effective ways of studying and reading can be recommended after a long day of classes, practicals, and other activities? Is setting alarms advisable?

First, an honest assessment: after a full day of classes and practicals, your cognitive capacity is genuinely reduced. Attempting to do your deepest, most demanding intellectual work at this point is fighting biology, and you will usually lose. The solution is not to push through blindly; it is to **match task type to cognitive state**.

Tiered study by energy level:

High energy (morning, before classes): This is your prime window. Use it for the hardest tasks; working through difficult problems, writing, synthesising new concepts, active self-testing.

Medium energy (between classes, early afternoon): Use this for reading, note consolidation, and reviewing material you have already encountered once.

Low energy (after a long day of practicals and activities): Use this for low-cognitive-load but still productive tasks:

- Reviewing flashcards (not creating them from scratch)
- Organising notes you took earlier
- Reading foundational or contextual material that does not require heavy analysis
- Planning tomorrow's study tasks

What to avoid in the low-energy window:

- Attempting to learn brand new complex concepts for the first time
- Writing essays or analytical pieces
- Doing timed practice problems requiring peak concentration

On the question of setting alarms:

Yes, with precision. A vague alarm that says "study time" is nearly useless. An alarm that says "15 minutes of flashcard review; Anatomy Chapter 4" is actionable. The alarm is a trigger; what it triggers must already be defined.

More importantly: **set an alarm to stop, not just to start.** One of the most effective practices for sustainable studying after a long day is the defined short block. Set a 30-minute timer, complete one specific task, stop. This is psychologically manageable when a three-hour session feels impossible, and it keeps the streak of daily engagement alive; which, over a semester, is worth more than occasional heroic study sessions.

One non-negotiable: sleep. Cognitive performance, memory consolidation, and emotional regulation all depend on it. Cutting sleep to study more is a trade that costs more than it gains. Seven to eight hours is not laziness; it is part of the system.

FAQ 4: Which works best; working hard or working smart?

This is a false choice, and accepting it as real is one of the most common reasons students plateau.

Here is the synthesis from all three books in one sentence: **Working smart determines the direction; working hard determines the distance.**

A student who works smart but does not put in sufficient hours will understand the method but lack the depth of practice to perform under exam conditions. A student who works hard using the wrong methods will accumulate hours without accumulating competence. Neither alone produces elite performance.

What working smart actually means; because the phrase has become so overused it has lost meaning:

Working smart means:

- Using retrieval practice instead of rereading
- Spacing study sessions instead of cramming
- Studying in high-intensity focused blocks rather than long distracted stretches
- Prioritising understanding of concepts over memorisation of facts
- Preparing for the *type* of thinking an exam requires, not just the content

What working hard actually means in this context:

It does not mean long hours. It means **sustained engagement with difficulty**. The performers who reach the highest levels are not those who work the most hours; they are those who consistently practice at the edge of their current competence, over a long period of time, without retreating to comfortable repetition.

The practical resolution:

Apply smart methods (retrieval practice, spaced repetition, active recall) with genuine effort and consistency across the full length of your semester; not just before assessments. That combination is what the research, the high-performing students Newport interviewed, and every elite performer Duckworth studied actually did. There is no shortcut that replaces it, and no amount of hours that compensates for studying in the wrong direction.

FAQ 5: Which other study techniques that work can you recommend based on research?

The following techniques are drawn from the strongest available research evidence. Each is presented with its mechanism; why it works, not just what it is; because understanding the mechanism helps you apply it correctly and adapt it to your context.

Technique 1: Retrieval Practice (Active Recall)

What it is: After studying material, close your notes and attempt to retrieve everything you can remember. Write it down. Then check what you missed.

Why it works: Every time you retrieve a memory, you strengthen the neural pathway to that memory. Recognition (seeing information and confirming you know it) does not produce this effect. Only generation (producing the answer from scratch) does.

How to apply it: After every lecture, before reviewing your notes, write a one-page reconstruction of what was covered. Use flashcards. Answer practice questions. Never let a study session end without a retrieval attempt.

Technique 2: Spaced Repetition

What it is: Instead of studying the same material in one long session, spread study of that material across multiple sessions separated by time; days or weeks.

Why it works: Memory consolidation requires time. When you return to material after a gap, you are forced to reconstruct it rather than simply refresh it from short-term memory. This reconstruction is what produces durable long-term retention.

How to apply it: Review new material 24 hours after first encounter, then again after one week, then again before the exam. Use a flashcard application like Anki, which automates this spacing automatically based on your performance.

Technique 3: Interleaved Practice

What it is: Instead of studying one topic exhaustively before moving to the next, mix different topics or problem types within a single study session.

Why it works: Blocked practice (studying one topic at a time) feels more productive because performance during the session is higher. But interleaving forces your brain to repeatedly identify *which* concept applies to *which* problem; exactly what an exam requires. It produces better long-term retention and transfer.

How to apply it: When doing practice problems, mix problems from different chapters rather than completing all problems from one chapter before moving on. In revision sessions, rotate between subjects rather than spending the entire session on one.

Technique 4: The Question-Evidence-Conclusion Note Framework

What it is: Structure all notes around three elements; what question is being answered, what evidence supports the answer, and what conclusion is drawn.

Why it works: It forces active processing during the lecture or reading rather than passive transcription. It also mirrors the structure of exam questions in most academic disciplines, so your notes are already formatted for retrieval.

How to apply it: Replace verbatim note-taking with this framework immediately. It requires practice, but within two to three weeks it becomes natural and your notes will be significantly more useful at revision time.

Technique 5: The Elaborative Interrogation Method

What it is: As you study, continuously ask "Why is this true?" and "How does this connect to what I already know?" Generate the answers yourself rather than accepting the text's explanation passively.

Why it works: New information is retained when it is connected to existing knowledge. The more connections you build between a new concept and things you already understand, the more retrieval pathways you create. This is why examples, analogies, and real-world applications improve retention.

How to apply it: At the end of every section you read, write two sentences: one explaining why the concept works the way it does, and one connecting it to something you already knew before this course.

Technique 6: Pre-Testing (Attempting Before Being Taught)

What it is: Before a lecture or before reading a chapter, attempt to answer questions about the material; even though you have not yet studied it.

Why it works: Struggling with a problem before being shown the solution dramatically improves retention of the solution when it arrives. The struggle creates a kind of cognitive gap that the brain is highly motivated to fill and remember.

How to apply it: Before each lecture, look at the topic and write down what you already know and what questions you expect it to answer. Before reading a chapter, attempt the end-of-chapter questions first. Your engagement with the actual content will be substantially deeper.

Final Note to All Students

Every technique listed above has one thing in common: it is harder and more uncomfortable in the moment than the alternatives. Rereading is easier than retrieval practice. Cramming feels more urgent than spaced repetition. Passive note-taking is less demanding than the Question-Evidence-Conclusion framework.

This discomfort is not a problem to overcome. It is the mechanism. The research across every study in *Make It Stick*, every student Newport interviewed, and every performer Duckworth examined confirms the same truth: **the difficulty you feel during correct studying is the sensation of your brain encoding information permanently.**

When studying feels hard, you are doing it right. When it feels easy and comfortable, you are almost certainly rehearsing what you already know rather than building what you need to know.

That distinction, internalised and acted on consistently, is the difference between studying hard and studying well.

Before You Leave

If you have read this far, then you are already different from most students.

Most people want better results, but very few are willing to study deeply enough to change the way they think, learn, and work. My hope is that this book becomes more than something you read. I hope it becomes something you practice.

I will continue creating resources for students who want to:

- move from average grades to academic excellence
- build stronger study systems
- learn faster and more effectively
- develop discipline and deep focus
- and compete at a higher academic level

This book is only the starting point.

You can find more materials, future releases, and additional resources from me through the links and platforms below.

<https://selar.com/m/david-itoya1>

Keep doing the hard things.

They change you.

— David Itoya

One More Thing Before You Go

If you enjoyed this book, there is a good chance you are the kind of person who cares deeply about growth — not just academically, but personally and professionally as well.

And while this book focused on helping you improve your academic performance, many of the same principles apply far beyond school:

- disciplined thinking
- systems over motivation
- deep work
- consistency
- learning valuable skills
- and building a life intentionally instead of randomly

That is part of why I created Scale Group.

Scale Group is a community and educational platform originally built for Nigerian entrepreneurs, creators, freelancers, and ambitious young people who want to grow financially and think at a higher level.

Now, you do **not** need to own a business to join.

In fact, many people inside are students — people trying to:

- build useful skills
- understand how money and business actually work
- become more disciplined
- meet growth-minded people
- and position themselves better for life after university

Inside the community, we share:

- practical frameworks
- business and growth lessons
- mindset shifts
- real-world case studies
- opportunities
- and conversations designed to help people think bigger and execute better

Most importantly, it is free to join.

So if you want to continue learning, growing, and surrounding yourself with people who are serious about becoming better, you are welcome to join us.

→ Join the Community here:

<https://chat.whatsapp.com/HLtfMHU31JXGCvVZUbLSoT>